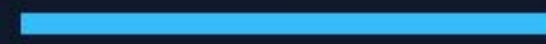


Iron Ore Trading

Quantitative Feature Engineering & Strategy Cards



IMPORT MARGIN • BASIS • MICROSTRUCTURE

Import Margin (PB Fines)

FORMULA

$$\text{Margin} = P_{spot} - [(P_{swap} - 0.92 \times C5 \times 0.088 + Prem) \times ExRate \times 1.13]$$

EXPLANATION

The difference between the landed cost and the spot price.

>0: Incentive to ship increases, future supply pressure rises.

<0: Reverse incentive.

INPUTS

- > P_spot (Spot Price)
- > P_swap (Swap 61)
- > C5 (Freight Cost)
- > Prem (Premium)
- > ExRate (CNY/USD)

SOURCE

Project Scope / Trade Accounting

Supply-Demand Gap

FORMULA

$$Gap_t = Supply_{China,t} - Demand_{Import,t}$$

EXPLANATION

Measures market tightness.

Gap > 0: Oversupply (Bearish).

Gap < 0: Undersupply (Bullish).

INPUTS

- › Shipments (AU + BR)
- › Domestic Supply
- › Import Demand Proxy (Hot Metal)

SOURCE

S&D Balance Framework

Basis Spread

FORMULA

$$Basis_t = P_{future,t} - (P_{spot,t} \times ExRate)$$

EXPLANATION

Reflects convergence and arbitrage space between futures and spot markets. Anomalous basis often indicates regression pressure or expectation gaps.

INPUTS

- › Futures Price
- › Spot Price
- › Exchange Rate

SOURCE

Arbitrage Theory

Inventory / Use Ratio

FORMULA

$$Ratio_t = \frac{PortInventory_t}{DailyHotMetalOutput_t}$$

EXPLANATION

Inventory buffer strength (Days of Usage). At low levels, restocking sensitivity rises, leading to higher price elasticity.

INPUTS

- › Port Inventory
- › Daily Hot Metal Output

SOURCE

Inventory Logic

In-Transit Inventory

FORMULA

$$InTransit_t = \sum_{i=1..N} Shipment_{t-i} - \sum_{i=1..N} Arrival_{t-i}$$

EXPLANATION

Hidden supply pressure from shipment to port arrival.
Used to forecast future arrivals and port congestion risks.

INPUTS

- › Shipments by Country
- › Arrivals
- › Window N (Days)

SOURCE

Logistics Accounting

Lump Premium Decomposition

FORMULA

$$Prem = CostSave + Scarcity + f(Supply, Demand)$$

EXPLANATION

Decomposes premium into "cost saving" (vs. Sinter) and "scarcity" components to enhance robustness and explainability.

INPUTS

- › Sinter Cost Proxy
- › S/D Metrics
- › Premium Data

SOURCE

Project / Industry Experience

Lump-Fines Spread

FORMULA

$$LFS_t = P_{lump,t} - P_{fines,t}$$

EXPLANATION

Reflects substitution logic in sintering/blending and steel mill preferences. Often leads structural demand changes.

INPUTS

- › Lump Ore Price
- › Fines Ore Price

SOURCE

Relative Pricing

Grade Spread (65% - 62%)

FORMULA

$$\text{Spread} = P_{65\%,t} - P_{62\%,t}$$

EXPLANATION

When steel margins are high, mills prefer high-grade ore for efficiency (Spread widens). Low margins compress the spread.

INPUTS

- › 65% Grade Price
- › 62% Grade Price

SOURCE

Grade Premium Logic

Steel / Ore Ratio

FORMULA

$$SOR_t = P_{rebar,t} / P_{ore,t}$$

EXPLANATION

A profit distribution indicator across the value chain. A very low ratio often signals that iron ore valuation is too high relative to steel.

INPUTS

- › Rebar Price
- › Iron Ore Price

SOURCE

Value Chain Pricing

Basis Residual (Sentiment)

FORMULA

$$residual_t = Basis_t - \mathbf{Model}(Fundamental_t)$$

EXPLANATION

After removing the explainable fundamental portion, the residual approximates speculative sentiment or noise.

INPUTS

- › Basis_t
- › Fitting Model Output

SOURCE

Feature Engineering

Lagged Returns

FORMULA

$$r_t = \ln(P_t / P_{t-1}); \text{Lag}_k = r_{t-k}$$

EXPLANATION

Foundational input for momentum and mean-reversion strategies.

INPUTS

- › Price Series P
- › Lag order k

SOURCE

Standard Practice

Rolling Mean & Volatility

FORMULA

$$MA_w = \text{mean}(P_{t-w..t}) \quad VOL_w = \text{std}(r_{t-w..t})$$

EXPLANATION

Smooths trends and quantifies risk/fluctuation. Essential for tree-based and deep learning models.

INPUTS

- › Price P or Return r
- › Window w

SOURCE

Standard Practice

Rolling Quantiles

FORMULA

$$Q_w^q(t) = \text{quantile}(r_{t-w..t}, q)$$

EXPLANATION

Captures tail risks and changes in distribution shape (skewness) over time.

INPUTS

- › Returns r
- › Window w
- › Quantile q

SOURCE

Standard Practice

Trend Slope

FORMULA

$$Slope_w = \text{OLS_slope}(P_{t-w..t} \text{ on time })$$

EXPLANATION

Measures the strength and direction of the trend within a window, reducing noise compared to raw returns.

INPUTS

- › Price P
- › Window w

SOURCE

Standard Practice

Intraday High-Low Range

FORMULA

$$Range_t = \ln(High_t / Low_t)$$

EXPLANATION

Uses intraday extremes to supplement volatility information, often superior to using Close-to-Close data alone.

INPUTS

- › High Price
- › Low Price

SOURCE

Volatility Features

Calendar & Seasonal Factors

FORMULA

{ DayOfWeek, MonthOfYear, HolidayFlag, DaysToExpiry }

EXPLANATION

Captures seasonality and institutional rhythms (Shipment schedules, Maintenance cycles, Policy windows).

INPUTS

- › Trading Calendar
- › Holiday Schedule

SOURCE

Standard Practice

Multi-scale MA

FORMULA

$$Vec_{MA} = [MA_{short}, MA_{mid}, MA_{long}]$$

EXPLANATION

Simultaneously inputs trends of different durations to enforce multi-cycle recognition in the model.

INPUTS

- › Price P
- › Set of windows W

SOURCE

Standard Practice

Supply-Demand Mismatch (Lagged)

FORMULA

$$GapLag_L(t) = Supply_{t-L} - Demand_t$$

EXPLANATION

Explicitly writes logistics time lags (e.g., shipping time) into features to improve interpretability and predictive power.

INPUTS

- › Supply Series
- › Demand Series
- › Lag L

SOURCE

Supply Chain Logic

VMD Decomposition Components

FORMULA

$$P_t \rightarrow \{ IMF_1, IMF_2, ..., IMF_k \}$$

EXPLANATION

Separates noise, cycle, and trend components using Variational Mode Decomposition for multi-scale parallel modeling.

INPUTS

- › Price P
- › VMD params (K, alpha)

SOURCE

Dragomiretskiy (2014)

Rolling Hurst Exponent

FORMULA

$$H_w = \text{Hurst}(P_{t-w..t})$$

EXPLANATION

$H > 0.5$: Trending behavior.

$H < 0.5$: Mean-reverting.

Used for regime perception and strategy switching.

INPUTS

- › Price P
- › Window w

SOURCE

Fractal Time Series

Dynamic Time Warping (DTW)

FORMULA

$$DTW(x,y) = \text{min_path} \sum d(x_i, y_j)$$

EXPLANATION

Aligns similar patterns of different speeds. Extracts "historical posterior" outcomes of similar past patterns as features.

INPUTS

- › Current Series x
- › Candidate History y

SOURCE

Time Series Similarity

Multi-Head Attention Weights

FORMULA

$$Att(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

EXPLANATION

Explicitly quantifies which historical time points the model focuses on. Used for interpretation and two-stage modeling.

INPUTS

- › Attention Matrix
- › Number of Heads

SOURCE

[Vaswani \(2017\) / Lim \(2021\)](#)

Head Time Scale Alignment

FORMULA

$H1: t-3..t-5$ (Short)

$H2: t-10..t-15$ (Mid)

$H3: \text{Event Mask}$

$H4: t-365 \pm k$ (Seasonal)

EXPLANATION

Aligns attention heads with business semantics: Short-term sentiment, mid-term lags, specific events, and seasonality.

INPUTS

- › Event List
- › Seasonality Window

SOURCE

Engineering Practice

Variable Selection Weights

FORMULA

$$w_t = \text{softmax}(VSN(x_t)); \quad \tilde{x}_t = \sum w_t \odot x_t$$

EXPLANATION

Calculates the weight of dominant factors for the current day, indicating regime or driver shifts.

INPUTS

› VSN Output Weights

SOURCE

Lim et al. (2021)

Dynamic Regime Weights

FORMULA

$$\hat{y}_t = \sum \alpha_{m,t} \cdot f_{m,t}; \quad \alpha_{m,t} = \mathbf{softmax}(g(z_t))$$

EXPLANATION

Automatically adjusts the contribution weight of different sub-models when the market state (Regime) changes.

INPUTS

- › Sub-model forecasts
- › State features z

SOURCE

Gated Ensemble

Two-Layer Dynamic Weights

FORMULA

$$\alpha = \lambda \cdot (RMSE^{-1}) + (1 - \lambda) \cdot \mathbf{softmax}(Regime)$$

EXPLANATION

Combines recent accuracy (short-term) with regime identification (long-term) to avoid rigid weighting schemes.

INPUTS

- › Rolling Error
- › Regime Label
- › Lambda

SOURCE

Engineering Practice

Fractional Differentiation

FORMULA

$$(1-B)^d = \sum w_k B^k; \quad w_k = -w_{k-1} \times \frac{d-k+1}{k}$$

EXPLANATION

Balances stationarity and memory preservation. Ideal for commodity price series with long memory.

INPUTS

- › Price P
- › Order d
- › Threshold τ

SOURCE

Lopez de Prado (AFML)

ADF Stationarity Test

FORMULA

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_i \delta_i \Delta y_{t-i} + \varepsilon_t$$

EXPLANATION

Tests for unit roots to avoid spurious regression. Used for robustness checks on features and targets.

INPUTS

- › Series y
- › Lags i

SOURCE

Dickey & Fuller (1979)

Granger Causality

FORMULA

$$y_t = \alpha + \sum \beta y_{t-i} + \sum \gamma x_{t-j} + \varepsilon_t$$

EXPLANATION

Verifies if X statistically precedes and aids in forecasting Y (e.g., Shipments leading Arrivals).

INPUTS

- › Series X, Y
- › Max Lag

SOURCE

Granger (1969)

Shannon Entropy

FORMULA

$$H = - \sum p_i \log(p_i)$$

EXPLANATION

Quantifies disorder and uncertainty changes. Helps identify environment shifts or "turning points".

INPUTS

- › Return Series
- › Binning Scheme

SOURCE

Shannon (1948)

Parkinson Volatility

FORMULA

$$\sigma_P = \text{sqrt}\left(\frac{1}{4 \ln 2} \times \left[\ln\left(\frac{\text{High}}{\text{Low}}\right) \right]^2\right)$$

EXPLANATION

Estimates volatility using intraday extremes. More sensitive to panic or extreme events than Close-to-Close volatility.

INPUTS

- › High Price
- › Low Price

SOURCE

Parkinson (1980)

VPIN (Informed Trading Prob)

FORMULA

$$VPIN \sim \frac{E[|V_{buy} - V_{sell}|]}{V_{total}}$$

EXPLANATION

Measures order flow toxicity and buy-sell imbalance.
Serves as an early warning for high volatility.

INPUTS

- › Tick/HFT Data
- › Volume Bars

SOURCE

Easley et al. (2012)

Cointegration Constraint

FORMULA

$$\text{Spread}_t = P1_t - \beta P2_t$$

Forecast clipped to: $[\mu - k\sigma, \mu + k\sigma]$

EXPLANATION

Prevents forecasts from violating common sense (e.g., High Grade < Low Grade Price) by enforcing statistical bounds.

INPUTS

- › Price Series 1 & 2
- › Bandwidth k

SOURCE

Engle & Granger (1987)

End of Deck

Quantitative Framework & Feature Engineering